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PI Name:	Roberts, Donna M.D.		
Project Title:	Safety and Efficacy of an Accelerated Protocol of Intermittent Theta Burst Transcranial Magnetic Stimulation (TMS) to Enhance Performance and Promote Resilience in Astronauts		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
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Space Biology Special Category:	None		
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Comments:			
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No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	George, Mark M.D. (Medical University of South Carolina) McTeague, Lisa Ph.D. (South Carolina Education Association)		
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Task Description:	Given the extreme conditions and stressful situations that will be encountered on exploration class missions, crews must demonstrate peak performance and resilience. Beyond medications, nutrition, and exercise, non-invasive brain stimulation is a method for maintaining and optimizing performance. Here we will develop and establish the efficacy of accelerated repetitive transcranial magnetic stimulation (rTMS) to left dorsolateral prefrontal cortex (dlPFC) as a non-pharmacological method for enhancing cognitive performance and resilience in high-performing healthy adults of astronaut age. First, we will establish an optimal dose of accelerated rTMS for enhancing neurocognitive performance and resilience that maximizes efficacy while minimizing burden. We will then examine the efficacy and safety of accelerated rTMS to left dlPFC for enhancing cognitive performance and resilience. Finally, we will evaluate the biological efficacy of TMS in a microgravity environment (parabolic flight).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	a. Our assessment of TMS resting motor threshold (rMT) in the zero gravity condition is an important baseline data point in understanding the response of the brain to acutely altered gravity and will facilitate the design of transcranial magnetic stimulation (TMS) treatment protocols for use on future spaceflight missions. Our results suggest that clinicians will need to consider altered corticospinal excitability during microgravity when designing TMS protocols. These results could also contribute to the understanding of altered inflight responses reported for psychotropic medications by astronauts and may be useful for physicians and pharmacists when prescribing these medications. b. No evidence of brain injury or structural brain changes were noted on magnetic resonance imaging (MRI) or decrements in cognitive performance following treatment supporting the safety of this intermittent theta-burst TMS protocol for enhancing cognitive function in high performing individuals such as astronauts. c. In a group of 40 healthy, well-educated participants, following a 1 week course of intermittent theta burst stimulation (iTBS) we found statistically significant increases in cognitive performance across several domains including: • The Spaceflight Cognitive Assessment Tool for Windows (WinSCAT) - symbol memory learning, sustained attention, and mathematical processing. • National Institutes of Health (NIH) Toolbox – fluid cognition composite, working memory, picture sequence memory, processing speed, inhibition control and attention. While a limitation of the study is the potential for practice effects, particularly with the WinSCAT which is known to be susceptible, to have contaminated the data, the NIH Toolbox has been well validated for longitudinal administration and shown to be relatively insensitive to practice effects. Also arguing against a practice effect in our data is the expected lack of change in crystallized cognition. Dose response curve analysis showed a non-significant trend towards enhanced performance at higher iTBS dose levels for the WinSCAT ICE composite score, symbol memory learning, and symbol memory recall. Although a dose-response curve plateau was not reached, our study showed that higher doses of rTMS are tolerable and safe and therefore future studies should include higher doses than those used in this study. Prior to this study, it was unknown what dose range could potentially have efficacy for enhancing neurocognitive performance in healthy, well-educated participants. This is the first such study undertaken in healthy adults and will serve as a guide for future studies. Taken together, as subjects required 12.6% less electromagnetism applied to the brain to cause thumb muscle activation during weightlessness compared to Earth gravity, perhaps TMS protocols such as the one used here for enhancing cognitive function and promoting resilience will be more effective at lower dose levels in microgravity than on Earth. It is interesting to speculate whether or not we would have reached a plateau in the dose-response curve had we used a 12% higher dose. Noninvasive brain stimulation is a novel method for enhancing performance across multiple cognitive domains and these data will be applicable to terrestrial patients as well as space travelers.
Task Progress:	1. Given the extreme conditions and stressful situations that will be encountered on exploration class missions, crews must demonstrate peak performance and resilience. Beyond medications, nutrition, and exercise, non-invasive brain stimulation is a method for maintaining and optimizing performance. Here, we developed and establish the efficacy of accelerated repetitive transcranial magnetic stimulation (rTMS) to left dorsolateral prefrontal cortex (dlPFC) as a non-pharmacological method for enhancing cognitive performance and resilience in high-performing healthy astronaut-like adults. First, in a group of 40 healthy adults, we established an optimal dose of accelerated rTMS for enhancing neurocognitive performance and resilience which maximized efficacy while minimizing burden. We also examined the safety of the accelerated rTMS protocol. Finally, we evaluated the biological efficacy of TMS in a microgravity environment (parabolic flight). This study represents the first randomized, double-blinded, controlled trial of accelerated rTMS to be performed in a healthy adult population in order to enhance cognitive performance and resilience. The study also provides important safety data for performing TMS in healthy populations. Finally, we provide data for translating Earth-based TMS protocols to the microgravity environment to facilitate the incorporation of TMS into future spaceflight missions. 2. Key findings: 2a. We developed and validated custom TMS helmets that allowed us to administer TMS in zero gravity with minimal training and high individual reliability. This TMS helmet method allows for reliable administration of TMS outside the laboratory setting with minimal training and may be useful in patient populations for clinical purposes as well. 2b. In order to consider the use of TMS as a countermeasure for spaceflight, the biological efficacy of TMS in a microgravity environment must be determined. Several TMS measures of cortical excitability have been described; however, the most basic is the TMS resting motor threshold (rMT). The rMT is believed to be an indicator of neuronal membrane excitability. Accurate determination of rMT is of utmost importance in both research and clinical studies as it is the unit most commonly used for rTMS dosing. Inaccurate estimation of rMT can lead to subject/patient over- or under-dosing. Here we have shown that rMT levels were 12.6% lower in zero gravity than they were pre- and post-flight in Earth gravity. These rMT changes were transient and did not persist after flight, and were not related to age, gender, or subjective emotional arousal at the time of data acquisition. 2c. Additional data will be provided. 3. Our assessment of rMT in the zero gravity condition is an important baseline data point in understanding the response of the brain to acutely altered gravity and will facilitate the design of TMS treatment protocols for use on future spaceflight missions. Our results suggest that clinicians will need to consider altered corticospinal excitability during

	microgravity when designing TMS protocols. These results could also contribute to the understanding of altered inflight responses reported for psychotropic medications by astronauts and may be useful for physicians and pharmacists when prescribing these medications.
Bibliography Type:	Description: (Last Updated: 01/15/2026)
Articles in Peer-reviewed Journals	Badran BW, Caulfield KA, Cox C, Lopez JW, Borckardt JJ, DeVries WH, Summers P, Kerns S, Hanlon CA, McTeague LM, George MS, Roberts DR. "Brain stimulation in zero gravity: transcranial magnetic stimulation (TMS) motor threshold decreases during zero gravity induced by parabolic flight." NPJ Microgravity. 2020 Sep 21;6:26. https://doi.org/10.1038/s41526-020-00116-6 . PMID: 33024819; PMCID: PMC7505837 , Sep-2020